

DESIGN

Three source CSVs → an 8-stage pipeline (90s, deterministic, no API key) → a DuckDB file plus a 31,795-vector index. Four tools over stdio:

- `get_schema` — columns, the values each takes, and an explicit list of what this data *cannot* answer.
- `run_sql` — one read-only SELECT, guarded.
- `find_similar_tickets` — nearest past tickets and their replies, or an explicit *no precedent*.
- `suggest_routing` — predicted queue and priority, with how far to trust it.

No LLM inside the server. Claude Code supplies the model; the server supplies grounded facts and refusals. It therefore needs no API key, and every tool is deterministic and testable.

WHY — MAKE THE MODEL UNABLE TO BE CONFIDENTLY WRONG

An LLM handed five weakly-related tickets still writes a fluent, confident reply. Models do not hedge by default. A tool's job is not to give the model access — it is to remove the ways it can be wrong.

Three instances, each measured rather than assumed:

- **It abstains.** The similarity floor is derived by sweeping thresholds on held-out data, not chosen. It accepts **97.5%** of real tickets and **0%** of 20 off-topic queries.
- **It says how much to trust itself.** Routing is **98.5%** accurate when a close precedent exists and **49%** when not. That spread is the product; the label alone is not.
- **The schema removes a trap.** Tags are stored positionally, so `GROUP BY tag_1` returns a confidently wrong number. Those columns are absent — the wrong query is unwritable, not discouraged.

WHAT THIS DATASET CANNOT SUPPORT

7.6% of answers are not support replies at all — customer text sitting in the answer column. There are no timestamps, no customer identifier, no money, no outcome field.

So this demonstrates a *mechanism*. It is not evidence about customers, and the page does not pretend otherwise.

BUSINESS IMPACT

The expensive thing in a support queue is elapsed time, not agent minutes. A first reply that says "please provide details" guarantees another exchange — the customer cannot tell what to send, and each leg is hours or days. `reply_state` marks which past replies named something fetchable, so a draft copies the ones that ended the round trip rather than the ones that caused it.

About half of all first replies are dead ends — they neither resolve the ticket nor say what to send. **49.8%** across all 39,739 answers, labelled by Haiku 4.5 for **\$9.53**; **55%** across the 40 I read by hand. Two independent estimates agreeing is the claim — its four-way taxonomy agrees with mine only **70%**, and why the collapse survives that is in `reports/labels.md`.

The delivery shape is **agent-to-agent handoff** — a front-line agent classifies and hands off to a specialist with the systems access. Calibrated confidence matters more there than accuracy: hand off above a threshold, escalate below it. A wrong handoff is recoverable; a confident wrong answer to a customer is not.

The transferable claim. CHEQ's Data & AI team will be asked to put internal data behind LLM tools. The bottleneck is not access, it is trust: point a model at raw warehouse tables and a stakeholder gets confidently wrong numbers, and the data team ends up trusted *less*. This repo is a working contract for doing that safely — the dataset is the substrate, not the point.

RISKS AND CONTROLS

Drafts a reply with no real precedent	Calibrated floor; returns <code>no_precedent</code>
Confidence misread on typed input	Two signals; guidance when they diverge
Silently wrong tag aggregates	Positional columns absent from schema
SQL reads local files / mutates	Capability removed, not denylisted
Memorised test set inflates scores	Deduplicated before splitting
German quietly worse than English	Every metric sliced by language

DELIBERATELY NOT BUILT

- **Vector database** — brute-force cosine over 40k×384 is sub-millisecond, and exact search adds no approximation error to confound the eval.
- **An LLM in the server** — the host already has one. Two would double cost and make the tools unevaluable.
- **A regex for the specific/vague split** — most replies open "could you please provide details" and then name the account number and the date. A rule keyed on that phrase inverts the answer on every one.
- **A reranker** — the ranking bias turned out to be a bug: subject-less tickets were 27% of top hits at a 10% base rate. Fixing that cost 0.006 macro-F1.
- **Fine-tuning** — the corpus is synthetic; a model would learn the template, not the domain.

PRODUCTION

- Corpus becomes Zendesk/Jira via Snowflake; the index rebuilds on an Airflow schedule and the abstention floor re-calibrates alongside it, since coverage grows.
- `draft_reply` moves server-side — a webhook has no host model — and posts to Slack under the ticket for a human to send.
- **The index sits outside the warehouse's policy boundary.** Snowflake row policies never see a vector search — embeddings are a copy of the data outside them. Re-gate separately: fetch permitted ids, mask, *then* search. The `queue` filter already works this way.
- Recency enters the ranking: a three-year-old precedent may describe a dead product.